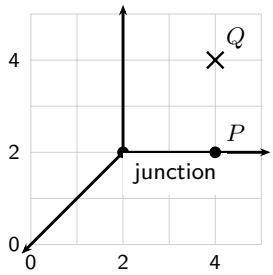
For a point (x, y), compare x, y, and 2. When the least number occurs at least twice, the point lies on this line.

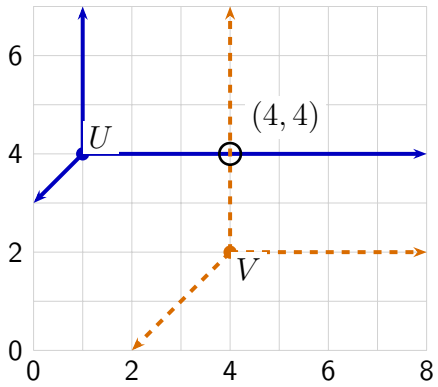


$P = (4, 2)$: $\quad 4,\ \underline{2},\ \underline{2}$
The least number is tied. P is on the line.

$Q = (4, 4)$: $\quad 4,\ 4,\ \underline{2}$
The least number occurs once. Q is off the line.

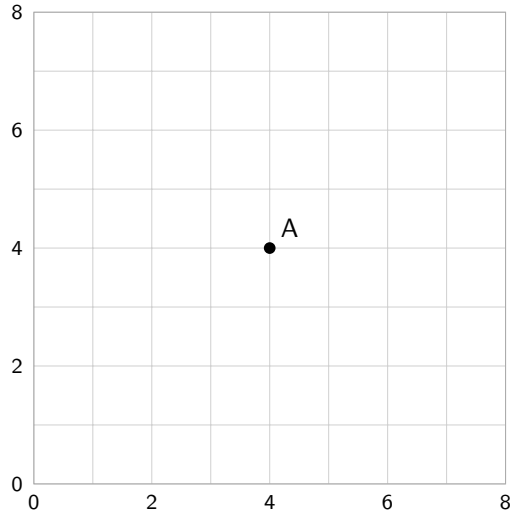
The arms point east, north, and diagonally southwest. They continue without end. Every line in this packet has this shape, moved without turning. Points between grid dots count too.

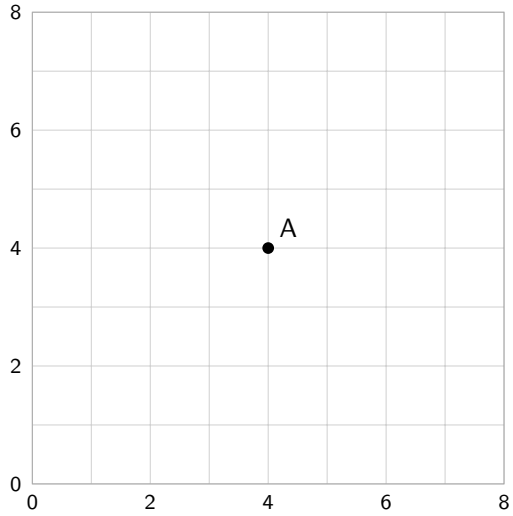
Here are two completed lines. Blue has junction $U = (1, 4)$; orange has junction $V = (4, 2)$. Their only shared point is circled: $(4, 4)$.

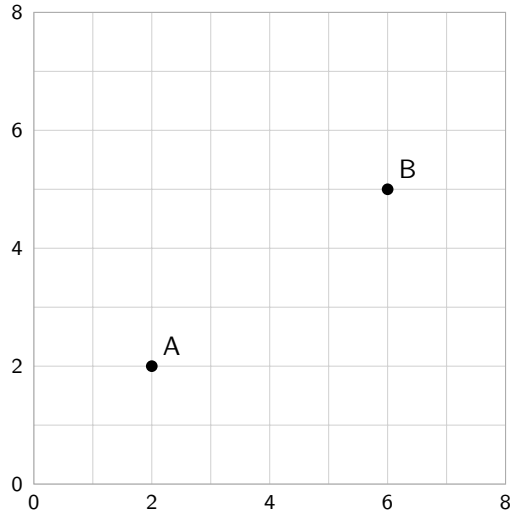


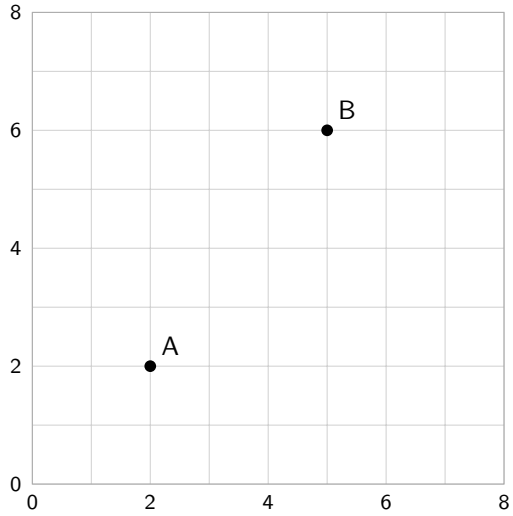
In the next problem, try to make two lines share just one point, then try to make two lines overlap along a whole stretch. A shared stretch includes every point between its grid dots.

Problem 1: Choose junction B on each top grid. Make the two lines share just one point on one top grid and a whole stretch on the other. On every grid, draw the lines with junctions A and B in different colours. Circle single shared points and trace the whole shared part heavily.

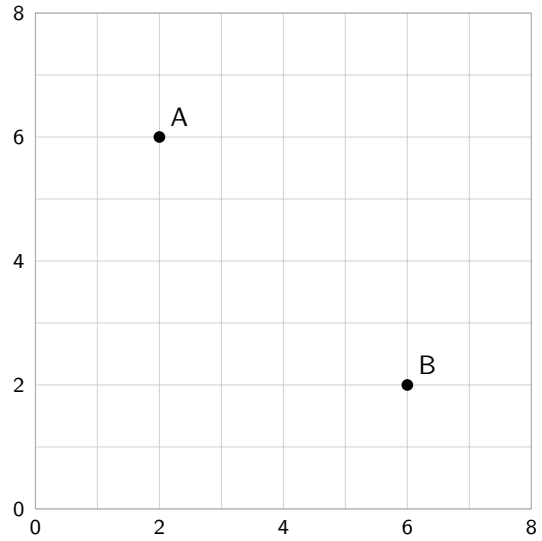


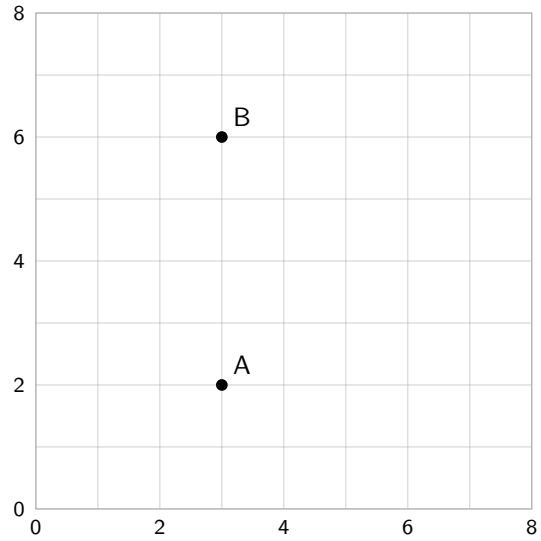


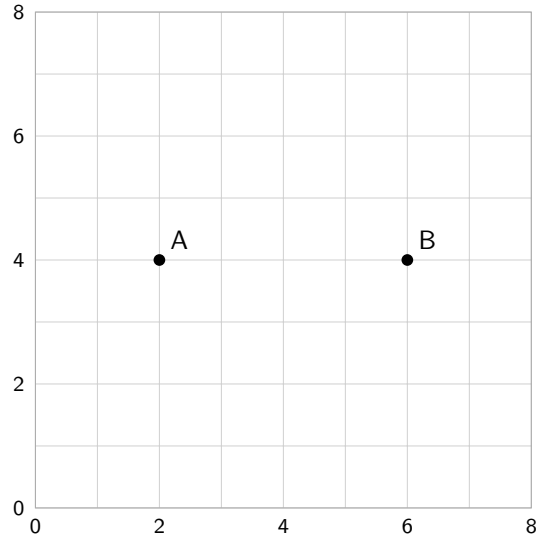


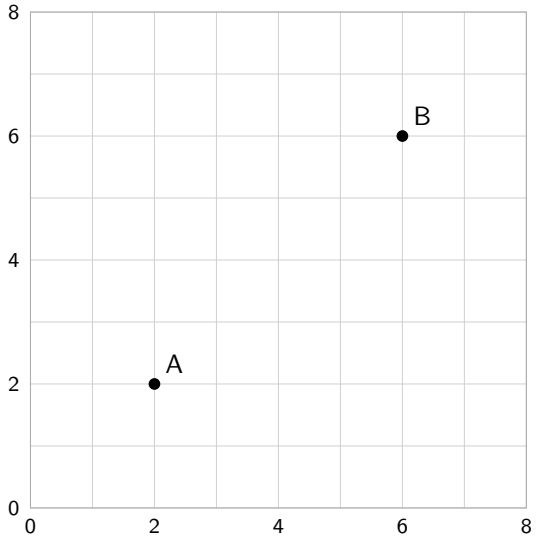


Problem 2: Draw the lines with junctions A and B on each grid. Mark every part they share. How many shared points does each pair have?

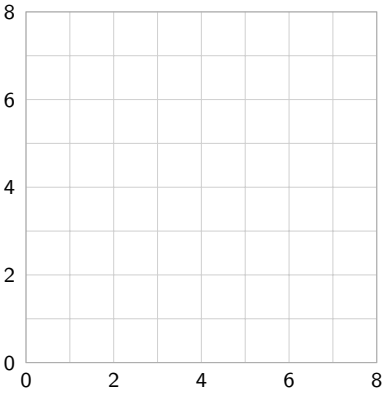




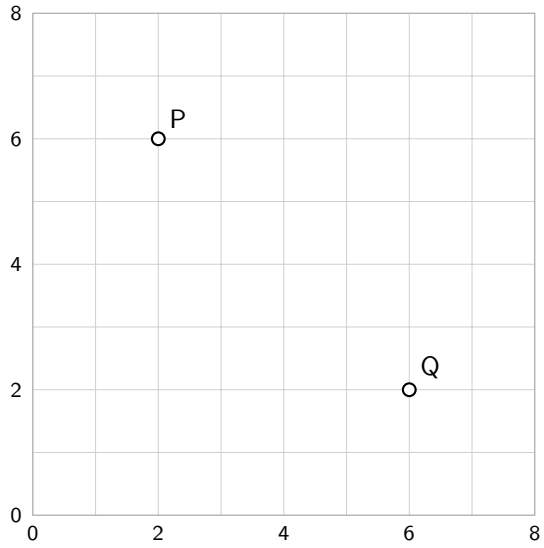


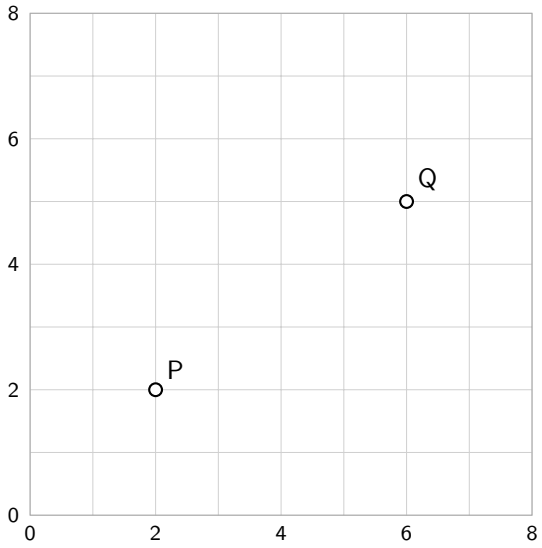


Problem 3: Can two lines with different junctions share exactly two points? Draw an example, or explain why this cannot happen.

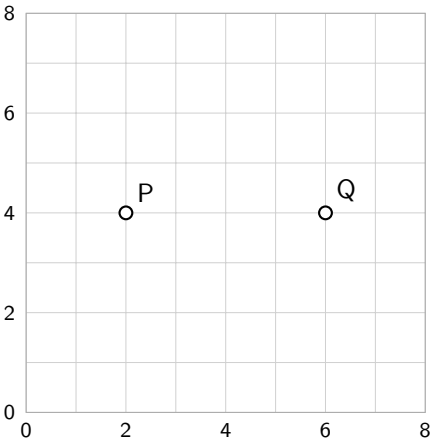


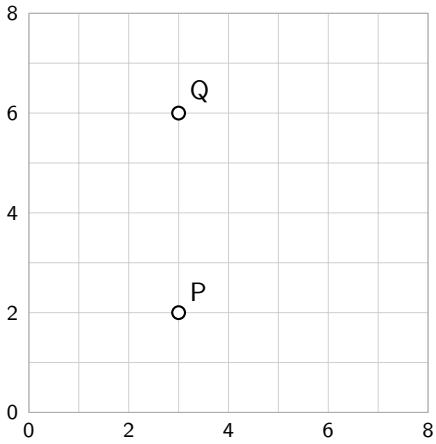
Problem 4: Draw a line through both points on each grid. Its junction may be anywhere, including between grid dots or outside the grid. Can you find a second line through the same two points?

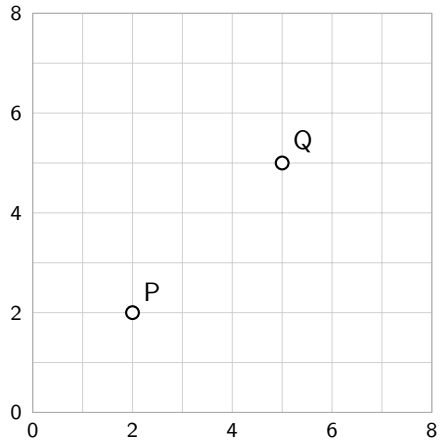




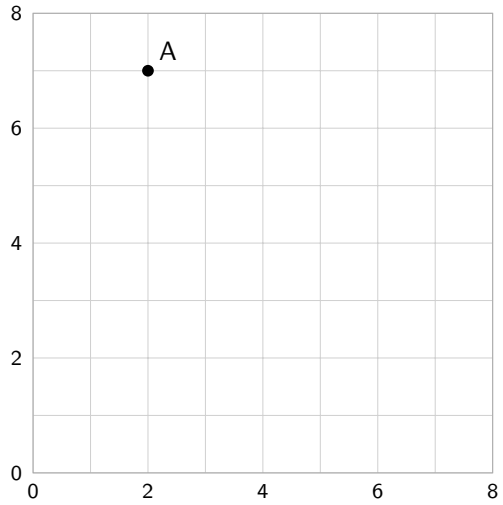
Problem 5: Find all possible junctions of a line through each pair of points.







Problem 6: Find where the lines with junctions $A = (2, 7)$ and $B = (10, 5)$ meet. Can any two lines miss each other? Explain your answer. Give a rule for when two lines share more than one point. Your rule must also work for junctions between grid dots or outside this page.



Problem 7: Find a rule that tells how many lines can pass through two different points. Explain why your rule works.

